MSc Data Science

DS601: Mathematical Foundations for Data Science

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Tutorial 1

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1. A class contains 60 students. Assume that a year has 365 days, birthdays are equally likely on all days, birthdays of different students are independent, and leap years are ignored. What is the probability that at least two students in the class have the same birthday? *(Class Activity)*

2. There are 30 students in a class. The teacher randomly selects three different students and assigns them the roles of President, Vice-President, and Secretary.

   a. Does the order matter?

   b. Is repetition allowed?

   c. How many possible role assignments are there?

   d. What is the probability that Student A is President, Student B is Vice-President, and Student C is Secretary?

3. Throwing a Four-Sided Die Twice

A fair four-sided die has the numbers 1, 2, 3, and 4. The die is thrown twice consecutively. Let X be the number obtained on the first throw and Y be the number obtained on the second throw.

   a. Write the sample space for this experiment.

   b. Find the probability that the sum $X + Y$ is odd.

   c. Find the probability of obtaining the numbers 1 and 4 in any order.

4. A committee of 8 members is to be selected from a pool of 18 candidates. The candidates are partitioned into four mutually exclusive categories containing 5, 5, 4, and 4 candidates, respectively. Determine the number of possible committees that intersect each category in a non-empty set.

5. A survey of 200 students found the following participation in clubs:

- Cricket (C) = 120
- Football (F) = 90
- Chess (H) = 80
- Cricket & Football = 50
- Cricket & Chess = 40
- Football & Chess = 30
- Cricket & Football & Chess = 20

Find the number of students in:

   a. At least one club.

   b. Exactly one club.

c. Exactly two clubs.

d. $U \setminus (C \cup F \cup H)$

6. A license plate consists of 3 letters followed by 3 digits. Letters are chosen from A-Z, and digits are chosen from 0-9. Repetition is allowed. How many possible license plates contain at least one vowel among the letters and at least one even digit among the digits?

7. How many 6-digit numbers satisfy all of the following conditions?

- The digits are distinct.
- The number is divisible by 5.
- The first digit is odd.